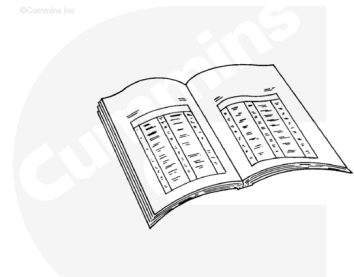


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

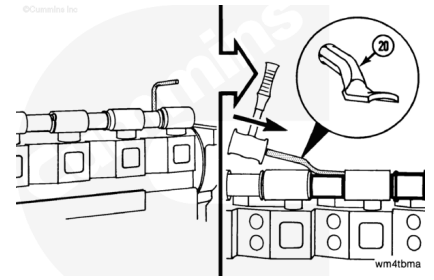
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html>).
- Remove the thermostat housing. Refer to Procedure 008-015 in Section 8 (</qs3/pubsys2/xml/en/procedures/18/18-008-015-tr.html>).
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/18/18-003-011-tr.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/18/18-003-009-tr.html>).
- Remove the crossheads and pushrods. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/18/18-004-014-tr.html>).

Remove

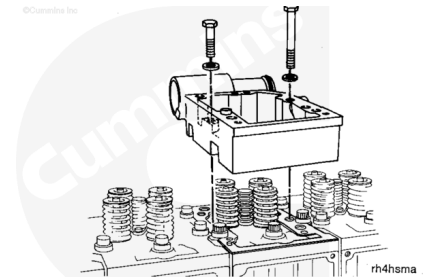
Using a hammer and water tube driver (20), Part Number ST-1319, drive the water tube toward the front of the engine until the back part of the tube clears the rocker lever housing.



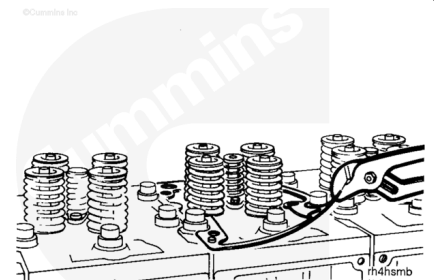
Some rocker lever housings have brackets, lifting eyes, and supports installed on the side of the housing. Record the location of the components so they can be installed in the correct location.

Remove the seven rocker lever housing mounting capscrews.

Remove the rocker lever housing.



Remove and discard the rocker lever housing gasket.



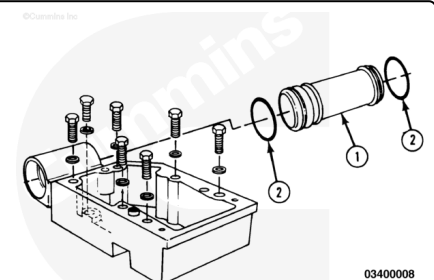
Disassemble

Remove the water transfer tube (1).

Remove the o-rings (2).

Discard the o-rings.

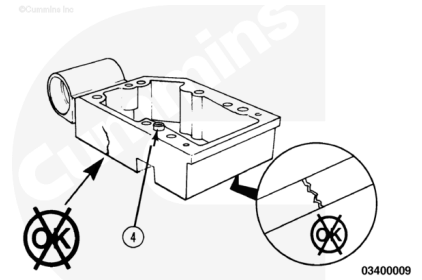
If the ring dowel is damaged, remove the dowel with the blind hole puller in the light duty puller kit, Part Number 3375784, or equivalent.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Use solvent or steam to clean the parts.

Inspect the ring dowel (4) for damage.

If the ring dowel is damaged it, **must** be replaced.

Use the dye crack penetrant method to inspect the rocker lever housing for cracks.

If the rocker lever housing is cracked, it **must** be replaced.

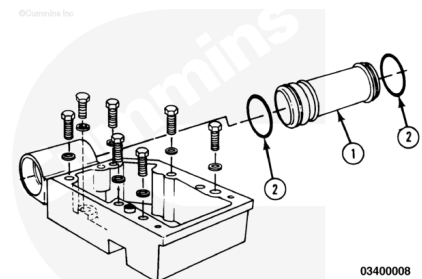
Assemble

Lubricate the o-rings (2) with vegetable oil.

Install the o-rings.

Install the water transfer tube (1).

Install the ring dowel if it was removed.



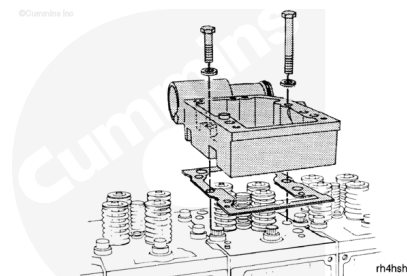
Install

Install the rocker lever housing gasket.

Align the rocker lever housing with the dowels and install the housing.

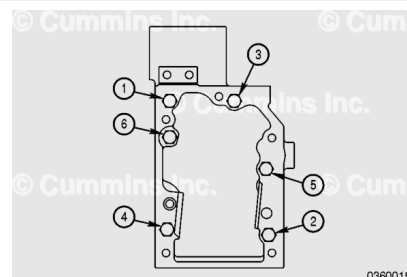
Check the threads of the rocker housing capscrews for damage. Check under the capscrew heads for cracks. Check for any deformation or necking of the capscrew.

Install the six capscrews.

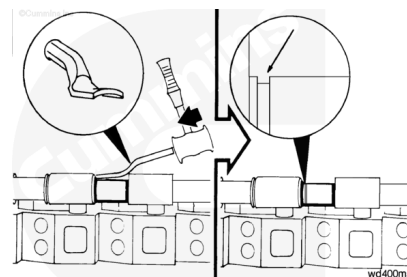


Tighten the capscrews in the sequence illustrated in the graphic.

Torque Value: 122 n•m [90 ft-lb]



Drive the water tube into the adjacent housing until the tube is approximately centered, with water tube driver, Part Number ST-1319, and a hammer.



Finishing Steps

- Install the push rods and crossheads. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/18/18-004-014-tr.html>).
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/18/18-003-009-tr.html>).
- Adjust the overhead. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/18/18-003-006-tr.html>) or Refer to Procedure 003-007 in Section 3 (</qs3/pubsys2/xml/en/procedures/18/18-003-007-tr.html>).
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/18/18->



003-011-tr.html).

- Install the thermostat housing. Refer to Procedure 008-015 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-015-tr.html).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html).
- Operate the engine to 70°C [160°F] minimum coolant temperature and check for leaks.